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Water quality detection device and cleaning method for sensor thereof

Abstract

A water quality detection device including a detection tank, a sensor, the cleaner and a processor is provided. The sensor is disposed on the detection tank and is configured to sense a to-be-detected liquid within the detection tank. The cleaner is configured to clean the sensor. The processor is electrically connected to the sensor and the cleaner and is configured to: execute an initialization procedure, which includes driving the sensor to sense the to-be-detected liquid to obtain a number of initial sensing values and calculating a threshold value according to the initial sensing values; drive the sensor to sense the to-be-detected liquid to obtain a sensing value of the to-be-detected liquid, and determine whether the sensing value of the to-be-detected liquid reaches the threshold value; drive the cleaner to operate when the sensing value of the to-be-detected liquid reaches the threshold value.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6779383	12/2003	Lizotte et al.	N/A	N/A
7542855	12/2008	Sihalla	N/A	N/A
8473262	12/2012	Farruggia et al.	N/A	N/A
9903793	12/2017	Meiman, Jr.	N/A	N/A
2003/0062262	12/2002	Mansouri	204/400	G01N 33/492
2014/0310895	12/2013	Chen	N/A	N/A
2014/0379302	12/2013	Park	N/A	N/A
2020/0165505	12/2019	Chen et al.	N/A	N/A
2021/0010989	12/2020	Yizhack et al.	N/A	N/A
2024/0177284	12/2023	Leport	N/A	G06T 7/194

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105618399	12/2016	CN	N/A
109029311	12/2017	CN	N/A
107110839	12/2018	CN	N/A
210318753	12/2019	CN	N/A
111595909	12/2019	CN	N/A
113607905	12/2020	CN	N/A
114236082	12/2021	CN	N/A
M582881	12/2018	TW	N/A
202004180	12/2019	TW	N/A
M604118	12/2019	TW	N/A
202238124	12/2021	TW	N/A

OTHER PUBLICATIONS

Taiwanese Office Action and Search Report for corresponding Taiwanese Application No. 111141408, dated Jul. 6, 2023. cited by applicant
Taiwanese Office Action and Search Report for Taiwanese Application No. 111141408, dated Mar. 8, 2024. cited by applicant
Cleary et al., “Development and Deployment of a Microfluidic Platform for Water Quality

Monitoring”, in Smart Sensors for Real-Time Water Quality Monitoring, Heidelberg, Berlin, 2013, pp. 125-148. cited by applicant
Delgado et al., “Antifouling Strategies for Sensors Used in Water Monitoring: Review and Future Perspectives”, Sensors 2021, vol. 21, 389, pp. 1-25. cited by applicant
Lizotte et al., “10 Tips to Prevent Biofouling on Water Quality Instruments”, YSI, Xylem, Inc., 2013, total 8 pages. cited by applicant

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Background/Summary

(1) This application claims the benefit of Taiwan application Serial No. 111141408, filed Oct. 31, 2022, the subject matter of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The disclosure relates in general to a water quality detection device and a cleaning method for a sensor thereof.

BACKGROUND

(3) After having been used over a period, most sensors of current water quality detection devices are susceptible to the deposition or attachment of impurities. For instance, if biofilm growth is not sensed and cleaned in time, sensing error will occur and the lifespan of the sensor will be affected. Since the water quality detection device normally needs to be monitored, maintained, and cleaned, extra labor is required, and the operation and maintenance cost of the water quality detection device will increase.

SUMMARY

(4) According to one embodiment of the disclosure, a water quality detection device. The water quality detection device includes a detection tank, a sensor, a cleaner and a processor. The sensor is disposed on the detection tank and is configured to sense a to-be-detected liquid within the detection tank. The cleaner is configured to clean the sensor. The processor is electrically connected to the sensor and the cleaner and is configured to: execute an initialization procedure, which includes driving the sensor to sense the to-be-detected liquid to obtain a number of initial sensing values and calculating a threshold value according to the initial sensing values; drive the sensor to sense the to-be-detected liquid to obtain a sensing value of the to-be-detected liquid, and determine whether the sensing value of the to-be-detected liquid reaches the threshold value; drive the cleaner to operate when the sensing value of the to-be-detected liquid reaches the threshold value.

(5) According to another embodiment of the disclosure, a cleaning method. The cleaning method is applicable to a water quality detection device, wherein the water quality detection device further includes a detection tank, a cleaner, a processor and a sensor. The cleaning method includes the following steps: executing at an initialization procedure, which includes driving the sensor to sense a to-be-detected liquid within the detection tank to obtain a plurality of initial sensing values and calculating a threshold value according to the initial sensing values; driving the sensor by the processor to sense the to-be-detected liquid to obtain a sensing value of the to-be-detected liquid, and determining by the processor whether the sensing value of the to-be-detected liquid reaches the threshold value; and driving the cleaner by the processor to operate when the sensing value of the to-be-detected liquid reaches the threshold value.

(6) The above and other aspects of the disclosure will become better understood with regard to the

following detailed description of the preferred but non-limiting embodiment (s). The following description is made with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a schematic diagram of a water quality detection device according to an embodiment of the disclosure.
- (2) FIG. 1B is a cross-sectional view of the water quality detection device of FIG. 1A along direction 1B-1B'.
- (3) FIG. 1C is a functional block diagram of the water quality detection device of FIG. 1A.
- (4) FIGS. 2A to 2B are schematic diagrams of a water quality detection device according to another embodiment of the disclosure.
- (5) FIG. 2C is a cross-sectional view of the water quality detection device of FIG. 2A along direction 2C-2C'.
- (6) FIG. 2D is a functional block diagram of the water quality detection device of FIG. 2A.
- (7) FIG. 3 is a flowchart of a cleaning method for the water quality detection device of FIG. 1A.
- (8) FIGS. 4A to 4B are flowchart of a cleaning method for the water quality detection device of FIG. 2A.
- (9) FIG. 5 is a relationship diagram of sensing value vs time for the sensor of FIG. 2A.

DETAILED DESCRIPTION

- (10) Refer to FIGS. 1A to 1C. FIG. 1A is a schematic diagram of a water quality detection device **100'** according to an embodiment of the present disclosure. FIG. 1B is a cross-sectional view of the water quality detection device **100'** of FIG. 1A along direction 1B-1B'. FIG. 1C is a functional block diagram of the water quality detection device **100'** of FIG. 1A.
- (11) As indicated in FIGS. 1A to 1C, the water quality detection device **100'** includes a detection tank **110**, a sensor **130**, a cleaner **140** and a processor **150**. The sensor **130** is disposed on the detection tank **110** and is configured to sense a to-be-detected liquid **L1** within the detection tank **110**. The cleaner **140** is configured to clean the sensor **130**. The processor **150** is electrically connected to the sensor **130** and the cleaner **140** and is configured to: execute an initialization procedure, which includes driving the sensor **130** to sense the to-be-detected liquid **L1'** (for example, a first liquid) to obtain a number of initial sensing values **R11** and calculating a threshold value **R11c** according to the initial sensing values **R11**; drive the sensor **130** to sense the to-be-detected liquid **L1** (for example, a second liquid) to obtain a sensing value of the to-be-detected liquid **R12** and determine whether the sensing value of the to-be-detected liquid **R12** reaches a threshold value **R11c**; and drive the cleaner **140** to operate when the sensing value of the to-be-detected liquid **R12** reaches the threshold value **R11c**. Thus, the water quality detection device **100'** could automatically determine the cleanliness of the sensor **130** according to the sensing value of the to-be-detected liquid **R12**. When the sensing value of the to-be-detected liquid **R12** reaches the threshold value **R11c**, this indicates that the cleanliness of the sensor **130** is abnormal and the accuracy of the sensing result may fail to meet the requirements. In such case, the cleaner **140** operates to clean the sensor **130**. Thus, the water quality detection device **100'** could sense the cleanliness of the sensor **130** and automatically clean the sensor **130** when the cleanliness of the sensor **130** is abnormal.
- (12) Refer to FIGS. 2A to 2D. FIGS. 2A to 2B are schematic diagrams of a water quality detection device **100** according to another embodiment of the present disclosure. FIG. 2C is a cross-sectional view of the water quality detection device **100** of FIG. 2A along direction 2C-2C'. FIG. 2D is a schematic diagram of the water quality detection device **100** of FIG. 2A. To avoid the diagrams becoming too complicated, the first pump **120A** and the second pump **120B** are only illustrated in

FIG. 2C and are not illustrated in FIGS. 2A to 2B.

(13) As indicated in FIGS. 2A to 2C, the water quality detection device **100** includes a detection tank **110**, a first pump **120A**, a second pump **120B**, a sensor **130**, a cleaner **140**, a processor **150**, an electromagnetic valve **160**. The sensor **130** is disposed on the detection tank **110** and is configured to sense the to-be-detected liquid **L1** within the detection tank **110**. The cleaner **140** is configured to clean the sensor **130** and includes a driver **170**. The processor **150** is electrically connected to the sensor **130** and the driver **170** of the cleaner **140** and is configured to: execute an initialization procedure, which includes: driving the sensor **130** to sense the to-be-detected liquid **L1'** to obtain a number of initial sensing values **R11** and calculating a threshold value **R11c** according to the initial sensing values **R11**; drive the sensor **130** to sense the to-be-detected liquid **L1** to obtain a sensing value of the to-be-detected liquid **R12**, and determine whether the sensing value of the to-be-detected liquid **R12** reaches the threshold value **R11c**; and drive the cleaner **140** to operate when the sensing value of the to-be-detected liquid **R12** reaches the threshold value **R11c**. Thus, the water quality detection device **100** could automatically determine the cleanliness of the sensor **130** according to the sensing value of the to-be-detected liquid **R12**. When the sensing value of the to-be-detected liquid **R12** reaches the threshold value **R11c**, this indicates that the cleanliness of the sensor **130** is abnormal, and the accuracy of the sensing result may fail to meet the requirements. In such case, the cleaner **140** operates to clean the sensor **130**. In other words, the water quality detection device **100** could sense the cleanliness of the sensor **130** and automatically clean the sensor **130** when the cleanliness of the sensor **130** is abnormal.

(14) The to-be-detected liquid **L1** and the to-be-detected liquid **L1'** each has a parameter, such as conductivity, pH value, dissolved oxygen, pH, or any other parameters representing other characteristics of the to-be-detected liquid **L1**. For instance, when the sensor **130** is a conductivity meter, the parameter is conductivity. Diverse types of parameters have their own ways in reflecting sensor abnormalities. (1). As the cleanliness sensed by the sensor **130** turns from a normal level to an abnormal level, the sensing value of the to-be-detected liquid **R12** increases to a threshold value **R11c**; or (2). As the cleanliness sensed by the sensor **130** turns from a normal level to an abnormal level, the sensing value of the to-be-detected liquid **R12** drop to the threshold value **R11c**. When the sensor is an electrode sensor, the sensing value of the to-be-detected liquid **R12** normally decreases to the threshold value **R11c**. Besides, the threshold value **R11c** could be a numeric value, but is not limited to a fixed value; the threshold value **R11c** could also be a numeric range. The to-be-detected liquid **L1** could be water from a river, a reservoir, a water tower, a ditch, or a fishpond, or could be wastewater, sewage water, or discharge water.

(15) As indicated in FIG. 2C, the first pump **120A** is electrically connected to the processor **150**. The first pump **120A** could transport a to-be-detected liquid **L1** to the detection tank **110**. The detection tank **110** has a liquid discharge port **110e**. The liquid within the detection tank **110** could be discharged off the detection tank **110** with the to-be-detected liquid **L1** (empty the detection tank **110**). The second pump **120B** is electrically connected to the processor **150**. The second pump **120B** could transport a standard liquid **L2** to the detection tank **110**. Before the standard liquid **L2** is transported to the detection tank **110**, when there is a liquid (such as the to-be-detected liquid **L2**) within the detection tank **110** already, the liquid (such as the to-be-detected liquid **L1**) could be discharged off the detection tank **110** via the liquid discharge port **110e** beforehand (empty the detection tank **110**).

(16) As indicated in FIG. 2C, the sensor **130** according to an embodiment of the present disclosure is exemplified by an electrode sensor. The sensor **130** includes a casing **131**, a first sensing electrode **132** and a second sensing electrode **133**, wherein the first sensing electrode **132** and the second sensing electrode **133** are disposed inside the casing **131**. The casing **131** has a notch **131a**, from which the first sensing electrode **132** and the second sensing electrode **133** are exposed to contact and sense the liquid within the detection tank **11**, such as the to-be-detected liquid **L1** or the standard liquid **L2**. Besides, one of the first sensing electrode **132** and the second sensing electrode

133 is a positive electrode and the other is a negative electrode. The processor **150**, electrically connected to the first sensing electrode **132** and the second sensing electrode **133**, receives sensing signals from the first sensing electrode **132** and the second sensing electrode **133** and analyzes or process sensing values.

(17) As indicated in FIG. 2C, the water quality detection device **100** further includes a carrier **135**. The carrier **135** is fixed on the detection tank **110** and has a through hole **135a**. The casing **131** of the sensor **130** could pass through the through hole **135a** and lean on the carrier **135** to fix the relative positions between the detection tank **110** and the sensor **130**. In an embodiment, the carrier **135** and the casing **131** could be locked, engaged, or temporarily bonded. Since the casing **131** of the sensor **130** could pass through the through hole **135a**, the first sensing electrode **132** and the second sensing electrode **133** arranged on the casing **131** face the cleaner **140** and could be cleaned thereby.

(18) As indicated in FIGS. 2A to 2C, the cleaner **140** includes a driver **170**, a first connection member **141**, a second connection member **142** and a brush **143**. The first connection member **141** has a first end **1411** and a second end **1412**. The first connection member **141** and the second connection member **142** could be separately manufactured and then could be screwed, engaged, or soldered temporarily or permanently. Or, the first connection member **141** and the second connection member **142** could be integrally formed in one piece without leaving any boundary surface or boundary layer on the junction. The brush **143** could be disposed on the second end **1412** of the first connection member **141**. One end of the second connection member **142** is connected to the first connection member **141**, and the other end is connected to the driver **170**. When the second connection member **142** rotates, the brush **143** connected to the first connection member **141** is driven to rotate accordingly (for instance, revolve around the X-axis) to clean the first sensing electrode **132** and the second sensing electrode **133** of the sensor **130**. The brush **143** may include a number of long strings for brushing the biofilm, impurities, or oil film off the sensing electrode. The long strings could be made of animal hairs or other materials, such as metal, plastic and/or rubber. Besides, the distance H1 between the sensing electrodes (the first sensing electrode and the second sensing electrode) and the first end **1411** is equivalent to the distance H2 between the first end **1411** and the second end **1412**. Thus, when the first connection member **141** rotates, the brush **143** located at the second end **1412** is driven to contact and clean the sensing electrodes.

(19) As indicated in FIGS. 2A to 2C, the processor **150** could be realized by a physical circuit formed using a semiconductor process. The processor **150** could be electrically connected to the first pump **120A**, the second pump **120B**, the sensor **130**, the electromagnetic valve **160**, and the driver **170** of the cleaner. The processor **150** could control the first pump **120A** and the second pump **120B** to transport a liquid to the detection tank **110**, receive the sensing signal of the sensor **130**, control the driver **170** to drive the cleaner **140**, and control the opening and closing of the electromagnetic valve **160**.

(20) As indicated in FIGS. 2A to 2C, the electromagnetic valve **160** is electrically connected to the processor **150**. The electromagnetic valve **160**, disposed on the bottom of the detection tank **110**, such as at the liquid discharge port **110e** on the bottom of the detection tank **110**, could control the opening and closing of the liquid discharge port **110e** to empty liquid and could close the bottom of the detection tank **110**. When the electromagnetic valve **160** is opened, the liquid within the detection tank **110** of the detection tank **110** is discharged off the detection tank **110** via the liquid discharge port **110e** and the electromagnetic valve **160**; when the electromagnetic valve **160** is driven to close the liquid discharge port **110e**, the bottom of the detection tank **110** could be closed to avoid the liquid within the detection tank **110** being discharged off the detection tank **110**.

(21) As indicated in FIGS. 2A to 2C, the driver **170** of the cleaner **140** includes a driving body **171** and a transmission shaft **172**. The transmission shaft **172** is connected to the driving body **171**, which drives the transmission shaft **172** to rotate (for instance, rotate around the X-axis). The second connection member **142** of the cleaner **140** is connected to the transmission shaft **172**,

wherein one end of the second connection member **142** of the cleaner **140** is connected to the transmission shaft **172** of the driver **170**. The second connection member **142** could be connected to the transmission shaft **172** of the driver **170**, so that the second connection member **142** could be driven by the driver **170** to rotate (for instance, rotate around the X-axis). Besides, the driving body **171**, being a waterproof driving module, could avoid the liquid within the detection tank **110** being discharged off the driving body **171**. In an embodiment, the driver **170** could be realized by a motor.

(22) As indicated in FIGS. 2A to 2C, the water quality detection device **100** further includes an adapter **173**; the driving body **171** is fixed on the adapter **173** and is therefore fixed on the detection tank **110** with the adapter **173**. The adapter **173** and the detection tank **110** could be screwed, engaged, or soldered temporarily or permanently. The transmission shaft **172** is protruded from the adapter **173** and faces the detection tank **110**. The detection tank **110** has a through hole **110a** interconnected with the detection tank **110**. The through hole **110a** allows the transmission shaft **172** to pass through, so that the cleaner **140** could be connected to the transmission shaft **172** within the detection tank **110**. During the assembly, firstly the adapter **173** could be arranged within the detection tank **110**, then the driver **170** of the cleaner **140** is arranged within the adapter **173**, wherein the transmission shaft **172** of the driver **170** passes through the through hole **110a** of the detection tank **110** to enter the detection tank **110**, then the second connection member **142** of the cleaner **140** is connected to the transmission shaft **172**. Besides, the water quality detection device **100** further includes a sealing member **115**, which could be interposed between the adapter **173** and the detection tank **110** to stop the liquid within the detection tank **110** leaking via the gap (if any) between the adapter **173** and the detection tank **110**.

(23) Details of the cleaning method for the water quality detection device **100'** according to an embodiment of the present disclosure are disclosed below.

(24) Refer to FIG. 3, a flowchart of a cleaning method for the water quality detection device **100'** of FIG. 1A is shown.

(25) In step **S110**, an initialization procedure is executed by the processor **150**. At the initialization procedure, the processor **150** drives the sensor **130** to sense the to-be-detected liquid **L1'** within the detection tank **110** to obtain a number of initial sensing values **R11** (step **S111**), and calculates the threshold value **R11c** according to the initial sensing values **R11** (step **S112**).

(26) In step **S120**, the processor **150** drives the sensor **130** to perform sensing on the to-be-detected liquid **L1** to obtain a sensing value of the to-be-detected liquid **R12**.

(27) In step **S130**, the processor **150** determines whether the sensing value of the to-be-detected liquid **R12** reaches the threshold value **R11c**. When the sensing value of the to-be-detected liquid **R12** reaches the threshold value **R11c**, the method proceeds to step **S140**. When the sensing value of the to-be-detected liquid **R12** does not reach the threshold value **R11c**, the method returns to step **S120**.

(28) In step **S140**, the processor **150** drives the cleaner **140** to operate.

(29) Details of the cleaning method for the water quality detection device **100** according to an embodiment of the present disclosure are disclosed below.

(30) Refer to FIGS. 4A to 4B and 5. FIGS. 4A to 4B are flowcharts of a cleaning method for the water quality detection device **100** of FIG. 2A. FIG. 5 is a relationship diagram of sensing value vs time for the sensor **130** of FIG. 2A.

(31) Firstly, the method begins at step **S210**, an initialization procedure including steps **S211** to **S212** is executed by the processor **150**.

(32) As indicated in FIG. 5, time interval $\Delta T1$ (for instance, a number of hours to a number of days) between time points **T0** to **T1** is the duration of executing step **S211** of the initialization procedure. In step **S211**, the sensor **130** performs sensing on the to-be-detected liquid **L1'** to obtain a number of initial sensing values **R11**. The number of items of initial sensing values **R11** depends on the sampling time. For instance, all the sensing values obtained by the sensor **130** during the

time interval $\Delta T1$ between time points $T0$ to $T1$ belong to initial sensing values $R11$. Or, the number of items of initial sensing values $R11$ is a predetermined number (for instance, tens to hundreds), and the length of the time interval $\Delta T1$ depends on the predetermined number or the sensing frequency.

(33) Each item of initial sensing values $R11$ is the sensing value of a renewed to-be-detected liquid $L1'$. In other words, to obtain each item of initial sensing values $R11$, the processor **150** opens the electromagnetic valve **160** to discharge the to-be-detected liquid $L1'$ that has been detected (to empty the detection tank **110**), then drives the electromagnetic valve **160** to close the liquid discharge port **110e** and drives the first pump **120A** to transport the to-be-detected liquid $L1'$ to the detection tank **110**, that is, to renew the to-be-detected liquid $L1'$ within the detection tank **110**. After the to-be-detected liquid $L1'$ is renewed, the processor **150** drives the sensor **130** to sense the to-be-detected liquid $L1'$ to obtain an item of initial sensing values $R11$. By the same analogy, the sensor **130** could obtain several items of initial sensing values $R11$.

(34) In step **S212**, the processor **150** obtains a threshold value $R11c$ of the initial sensing values $R11$ according to the initial sensing values $R11$. In a situation, the threshold value $R11c$ is where the cleanliness of the sensing electrode of the sensor **130** turns from a normal level to an abnormal level. As the cleanliness of the sensing electrode turns from a normal level to an abnormal level, when the initial sensing value $R11$ decreases, the threshold value $R11c$ could be the minimum of the initial sensing values $R11$ or a value lower than the average value of the initial sensing values $R11$. In another situation, as the cleanliness of the sensing electrode turns from a normal level to an abnormal level, when the initial sensing values $R11$ increase, the threshold value $R11c$ could be the maximum of the initial sensing values $R11$ or a value higher than the average of the initial sensing values $R11$.

(35) In the embodiment, as indicated in FIG. 5, the threshold value $R11c$ is the minimum of the initial sensing values $R11$ (such as a sensing value within the time interval $\Delta T1$).

(36) After executing the initialization procedure to obtain the threshold value $R11c$, the processor **150** then executes a measurement procedure, which includes steps **S214** to **S218**. In the embodiment, as indicated in FIG. 5, the time interval $\Delta T2$ between time points $T1$ to $T2$ is the period of steps **S214** to **S218**.

(37) In step **S214**, the initial value of the cleaning count n is exemplarily set to 0 by the processor **150**.

(38) In step **S215**, the processor **150** drives the electromagnetic valve **160** to open the liquid discharge port **110e** to empty the detection tank **110**, then drives the electromagnetic valve **160** to close the liquid discharge port **110e**, and the first pump **120A** to transport the to-be-detected liquid $L1$ to the detection tank **110**.

(39) In step **S216**, the sensor **130** is driven by the processor **150** to sense the to-be-detected liquid $L1$ to obtain a sensing value of the to-be-detected liquid $R12$.

(40) In step **S218**, the processor **150** determines whether the sensing value of the to-be-detected liquid $R12$ reaches the threshold value $R11c$. When the processor **150** determines that the sensing value of the to-be-detected liquid $R12$ does not reach the threshold value $R11c$, the method proceeds to step **S216**. In step **S216**, the sensor **130** is driven to sense the next preparation of to-be-detected liquid $L1$ to obtain the next sensing value of the to-be-detected liquid $R12$. By the same analogy, the processor **150** could obtain several sensing values of the to-be-detected liquid $R12$ via the sensor **130**.

(41) Each sensing value of the to-be-detected liquid $R12$ is the sensing value of a renewed to-be-detected liquid $L1$ within the detection tank **110**. To put it in greater details, before the next sensing value of the to-be-detected liquid $R12$ is obtained, the processor **150** opens the electromagnetic valve **160** to discharge the to-be-detected liquid $L1$ within the detection tank **110** that has been detected, that is, to empty the detection tank **110**, then closes the electromagnetic valve **160**. Then the processor **150** drives the first pump **120A** to transport the to-be-detected liquid $L1$ to the

detection tank **110** and drives the sensor **130** to sense the renewed to-be-detected liquid **L1** to obtain the next sensing value of the to-be-detected liquid **R12**.

(42) On the other hand, when the processor **150** determines that the sensing value of the to-be-detected liquid **R12** reaches the threshold value **R11c**, the processor **150** executes steps **S220** to **S232**. As indicated in FIG. 5, the time interval $\Delta T3$ between time points **T2** to **T3** is the period of steps **S220** to **S232**.

(43) In step **S220**, the processor **150** drives the cleaner **140** to execute a cleaning procedure for cleaning the sensor **130**. As indicated in FIGS. 2A to 2B, the processor **150** controls the transmission shaft **172** of the driver **170** of the cleaner **140** to rotate, so that the first connection member **141** and the second connection member **142** of the cleaner **140** could drive the brush **143** to rotate to clean the sensing electrode of the sensor **130**. The processor **150** drives the cleaner **140** to execute a cleaning procedure. At the cleaning procedure, the cleaner **140** continuously cleans the sensor **130** for a cleaning time, then stops. The cleaning time is within a range of 30 to 60 seconds.

(44) In step **S222**, the cleaning count n is increased by 1 by the processor **150**, for instance, $n=n+1$.

(45) In step **S224**, the processor **150** drives the electromagnetic valve **160** to open the liquid discharge port **110e** to empty the detection tank **110**, then drives the electromagnetic valve **160** to close the liquid discharge port **110e** and drives the first pump **120A** to transport another (or the next) preparation of to-be-detected liquid **L1** to the detection tank **110**, that is, to renew the to-be-detected liquid **L1** within the detection tank **110**.

(46) In step **S226**, the processor **150** drives the sensor **130** to sense the another to-be-detected liquid **L1** to obtain another (or the next) sensing value of the to-be-detected liquid **R12**.

(47) In step **S228**, the processor **150** determines whether the another sensing value of the to-be-detected liquid **R12** obtained in step **S226** still reaches the threshold value **R11c**. If so, the method proceeds to step **S232**. If no, this indicates that the cleanliness of the sensing electrode of the sensor **130** is already back to the normal level after cleaning (for instance, at the time point **T3'** of FIG. 5), then the method proceeds to step **S230**, the processor **150** resets the cleaning count n to zero (for instance, $n=0$) and the process again proceeds to step **S216** to sense the next preparation of to-be-detected liquid **L1**. As disclosed above, before the next preparation of to-be-detected liquid **L1** is sensed, the processor **150** drives the electromagnetic valve **160** to open to discharge the to-be-detected liquid **L1** that has been detected, then closes the electromagnetic valve **160** and drives the first pump **120A** to transport a to-be-detected liquid **L1** to the detection tank **110**, and drives the sensor **130** to sense the next preparation of to-be-detected liquid **L1** to obtain a next sensing value of the to-be-detected liquid **R12**.

(48) In step **S232**, the processor **150** determines whether the cleaning count n reaches the predetermined number **N**, that is, the processor **150** determines the number of times, denoted by n , for which the cleaner **140** cleans the sensor **130** reaches the predetermined number **N**. The predetermined number **N** is a numeric value greater than or equivalent to 1, such as any positive integer within a range of 1 to 30 or even larger. When it is determined that the cleaning count n reaches the predetermined count **N**, the method proceeds to step **S234**; when it is determined that the cleaning count n does not reach the predetermined count **N**, the method proceeds to step **S220**, the cleaner **140** again cleans the sensor **130**.

(49) When the cleaning count n reaches the predetermined number **N** and the sensing value of the to-be-detected liquid **R12s** still reaches the threshold value **R11c**, a sensor failure determination procedure (steps **S234** to **S240**) is executed to determine whether the sensor **130** has failed. Under such circumstance, the cleaning of the sensor **130** does not help much to improve the cleanliness of the sensor **130** because the sensor **130** may have failed. If the sensor **130** fails, despite that the cleanliness of the sensing electrode of the sensor **130** is normal, the sensing value of the to-be-detected liquid **R12** still may reach the threshold value **R11c**.

(50) Details of the sensor failure determination procedure (steps **S234** to **S240**) are disclosed below.

(51) In step **S234**, the processor **150** drives the electromagnetic valve **160** to open the liquid

discharge port **110e** to empty the detection tank **110**, then drives the electromagnetic valve **160** to close the liquid discharge port **110e** and the second pump **120B** to transport a standard liquid L2 to the detection tank **110**.

(52) In step **S236**, the processor **150** drives the sensor **130** to sense the standard liquid L2 within the detection tank **110** to obtain a sensing value **R21** of the standard liquid L1, wherein the parameter of the standard liquid L2 and the parameter of the to-be-detected liquid L1 belong to the same type. For instance, when the parameter of the to-be-detected liquid L1 is conductivity, the parameter of the standard liquid L2 is also conductivity.

(53) In step **S238**, the processor **150** determines whether the error between the sensing value **R21** of the standard liquid L1 and the standard value **R22** is equivalent to or larger than an allowable error value. When the error is not equivalent to or larger than the allowable error value, the method proceeds to step **S210** (the initialization procedure), that is, the threshold value **R11c** is redefined; when the error is equivalent to or larger than the allowable error value, the method proceeds to step **S240**.

(54) In other words, when the error between the sensing value **R21** of the standard liquid L1 and the standard value **R22** is not larger than the allowable error value, this indicates the result “the cleaning count *n* reaches the predetermined number *N* but the sensing value of the to-be-detected liquid **R12** still reaches the threshold value **R11c**” is not caused by the failure of the sensor **130**, that is, the sensor **130** still could be used but the threshold value **R11c** needs to be redefined, therefore step **S210** is executed again. The “allowable error value” is in a range of 1% to 3%; however, the allowable error value could be smaller than 1% or larger than 3%.

(55) In step **S240**, when the error between the sensing value **R21** and the standard value **R22** is lower than the allowable error value, the processor **150** generates a sensor maintenance signal **S1** to prompt the user to maintain the sensor **130**. When the error between the sensing value **R21** and the standard value **R22** is lower than the allowable error value, this indicates that the result “the cleaning count *n* reaches the predetermined number *N* but the sensing value of the to-be-detected liquid **R12** still reaches the threshold value **R11c**” could be caused by the failure of the sensor **130**, and the sensor **130** could no longer be used. Thus, the user could replace the sensor **130** according to the sensor maintenance signal **S1**.

(56) To summarize, the water quality detection device and the cleaning method for a sensor thereof of the present disclosure are capable of automatically sensing the cleanliness of the sensor and automatically cleaning the sensor (self-cleaning) when the cleanliness of the sensor is abnormal, so that the labor cost of maintenance could be reduced and the sensing accuracy could last for a longer period of time.

(57) It will be apparent to those skilled in the art that various modifications and variations could be made to the disclosed embodiments. It is intended that the specification and examples be considered as exemplary only, with a true scope of the disclosure being indicated by the following claims and their equivalents.

Claims

1. A water quality detection device, comprising: a detection tank; a sensor disposed on the detection tank and configured to sense a first liquid within the detection tank; a cleaner configured to clean the sensor; a first pump; and a processor electrically connected to the sensor, the cleaner and the first pump, and configured to: execute an initialization procedure, which comprises driving the sensor to sense the first liquid to obtain a plurality of initial sensing values and calculating a threshold value according to the initial sensing values; drive the first pump to transport a second liquid to the detection tank after emptying the detection tank of the first liquid; drive the sensor to sense the second liquid to obtain a sensing value of the second liquid, and determine whether the sensing value of the second liquid reaches the threshold value; and drive the cleaner to operate

when the sensing value of the second liquid reaches the threshold value.

2. The water quality detection device according to claim 1, further comprising an electromagnetic valve electrically connected to the processor, wherein the detection tank has a liquid discharge port, the electromagnetic valve is configured to control the opening and closing of the liquid discharge port to empty the detection tank.
3. The water quality detection device according to claim 2, further comprising a second pump electrically connected to the processor, wherein after the processor drives the cleaner to complete a cleaning procedure, the processor is further configured to: drive the electromagnetic valve to open the liquid discharge port to empty the detection tank, and then drive the electromagnetic valve to close the liquid discharge port and the first pump to transport the second liquid to the detection tank; drive the sensor to sense the second liquid to obtain another sensing value of the second liquid; determine whether the another sensing value of the second liquid reaches the threshold value; determine whether a cleaning count reaches a predetermined number when the another sensing value of the second liquid reaches the threshold value; and when the cleaning count reaches the predetermined number, the processor: drives the electromagnetic valve to open the liquid discharge port to empty the detection tank, and then drives the electromagnetic valve to closes the liquid discharge port and the second pump to transport a standard liquid to the detection tank; drives the sensor to sense the standard liquid to obtain a sensing value of the standard liquid, and determines whether an error between a sensing value of the standard liquid and a standard value of the standard liquid is equivalent to or larger than an allowable error value; and executes the initialization procedure again when the error between the sensing value of the standard liquid and the standard value is not larger than the allowable error value.
4. The water quality detection device according to claim 3, wherein the processor is further configured to: generate a sensor maintenance signal when the error between the sensing value of the standard liquid and the standard value is equivalent to or larger than the allowable error value.
5. The water quality detection device according to claim 1, wherein the processor is further configured to: determine whether the sensing value of the second liquid is equivalent to or lower than the threshold value; and drive the cleaner to execute a cleaning procedure when the sensing value of the second liquid is equivalent to or lower than the threshold value.
6. The water quality detection device according to claim 1, wherein the processor is configured to drive the cleaner to execute a cleaning procedure, and after the cleaning procedure is completed, the processor is further configured to: increase a cleaning count by 1, and drive the sensor to sense the second liquid to obtain another sensing value of the second liquid; determine whether the another sensing value of the second liquid reaches the threshold value; determine whether the cleaning count reaches a predetermined number when the another sensing value of the second liquid reaches the threshold value; and drive the cleaner to execute the cleaning procedure, and increase the cleaning count by 1 when the cleaning count does not reach the predetermined number.
7. The water quality detection device according to claim 6, further comprising an electromagnetic valve electrically connected to the processor, wherein the detection tank has a liquid discharge port, the electromagnetic valve is configured to control the opening and closing of the liquid discharge port to empty the detection tank, and the processor is further configured to: drive the electromagnetic valve to open the liquid discharge port to empty the detection tank, and then drive the electromagnetic valve to close the liquid discharge port and the first pump to transport the second liquid to the detection tank before the sensor executes sensing to obtain the another sensing value of the second liquid; and reset the cleaning count to zero when the another sensing value of the second liquid does not reach the threshold value.
8. The water quality detection device according to claim 1, wherein the cleaner comprises: a driver comprising a transmission shaft; a connection member having a first end and a second end, wherein the first end is connected to the driver; and a brush disposed on the second end of the connection member.

9. The water quality detection device according to claim 8, wherein the sensor comprises a sensing electrode, and a distance between the sensing electrode and the first end is equivalent to a distance between the first end and the second end.

10. A cleaning method applicable to a water quality detection device, wherein the water quality detection device further comprises a detection tank, a cleaner, a processor, a sensor and a first pump, and the cleaning method comprises: executing an initialization procedure, which comprises driving the sensor to sense a first liquid within the detection tank to obtain a plurality of initial sensing values and calculating a threshold value according to the initial sensing values; drive the first pump to transport a second liquid to the detection tank after emptying the detection tank of the first liquid; and driving the sensor by the processor to sense the second liquid to obtain a sensing value of the second liquid, and determining by the processor whether the sensing value of the second liquid reaches the threshold value; driving the cleaner by the processor to operate when the sensing value of the second liquid reaches the threshold value.

11. The cleaning method according to claim 10, wherein the water quality detection device further comprises an electromagnetic valve electrically connected to the processor, and the detection tank has a liquid discharge port; the cleaning method further comprises: controlling the electromagnetic valve by the processor to open the liquid discharge port to empty the detection tank.

12. The cleaning method according to claim 11, wherein the water quality detection device further comprises a second pump electrically connected to the processor; after the processor drives the cleaner to complete the cleaning procedure, the cleaning method further comprises: driving the electromagnetic valve to open the liquid discharge port to empty the detection tank, and then driving the electromagnetic valve to close the liquid discharge port and the first pump to transport the second liquid to the detection tank by the processor; driving the sensor by the processor to sense the second liquid to obtain another sensing value of the second liquid; determining by the processor whether the another sensing value of the second liquid reaches the threshold value; determining by the processor whether a cleaning count reaches a predetermined number when the another sensing value of the second liquid reaches the threshold value; and when the cleaning count reaches the predetermined number, the cleaning method further comprises: driving the electromagnetic valve to open the liquid discharge port to empty the detection tank, and then driving the electromagnetic valve to close the liquid discharge port and the second pump to transport a standard liquid to the detection tank by the processor; driving the sensor to sense the standard liquid to obtain a sensing value of the standard liquid and determine by the processor whether an error between a sensing value of the standard liquid and a standard value of the standard liquid is equivalent to or larger than an allowable error value; and executing the initialization procedure by the processor again when the error between the sensing value of the standard liquid and the standard value is not larger than the allowable error value.

13. The cleaning method according to claim 12, further comprising: generating a sensor maintenance signal by the processor when the error between the sensing value of the standard liquid and the standard value is equivalent to or larger than the allowable error value.

14. The cleaning method according to claim 10, further comprising: determining by the processor whether the sensing value of the second liquid is equivalent to or lower than the threshold value; and driving the cleaner to execute a cleaning procedure by the processor when the sensing value of the second liquid is equivalent to or lower than the threshold value.

15. The cleaning method according to claim 10, further comprising: driving the cleaner by the processor to execute a cleaning procedure; after the cleaner completes the cleaning procedure, the cleaning method further comprises: increasing a cleaning count by 1 and driving the sensor to sense the second liquid by the processor to obtain another sensing value of the second liquid; determining by the processor whether the another sensing value of the second liquid reaches the threshold value; determining by the processor whether the cleaning count reaches a predetermined number when the another sensing value of the second liquid reaches the threshold value; and driving the

cleaner to execute the cleaning procedure by the processor and increasing the cleaning count by 1 when the cleaning count does not reach the predetermined number.

16. The cleaning method according to claim 15, wherein the water quality detection device further comprises an electromagnetic valve electrically connected to the processor, and the detection tank has a liquid discharge port; the cleaning method further comprises: driving the electromagnetic valve to open the liquid discharge port to empty the detection tank, and then driving the electromagnetic valve to close the liquid discharge port and the first pump to transport the second liquid to the detection tank by the processor before the sensor executes sensing to obtain the another sensing value of the second liquid; and resetting the cleaning count to zero by the processor when the another sensing value of the second liquid does not reach the threshold value.
